D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

WHO 2022 classification of Myeloid malignancies and describe in detail diagnostic investigations, molecular testing, ELN 2022 risk stratification and management of Acute Myeloid Leukemia (AML).

February 2023 | DM 0223 | Q2 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction & exam framing

WHO 2022 (5th edition) reorganised myeloid neoplasms around genetics-first definitions, narrowing morphology-only categories and aligning partly with ICC 2022. AML diagnosis, ELN 2022 genetic risk, measurable residual disease (MRD), and genomically targeted therapy (FLT3, IDH1/2, menin pathway) dominate modern management.

15-mark skeleton: WHO myeloid map → AML diagnostic work-up → molecular panel → ELN 2022 risk table → fit vs unfit algorithms with doses → landmark trials → NCCN AML v5.2026 (ziftomenib NPM1 R/R, revumenib, HMA+ven, FLT3) → pearls.

- Separate APL (PML::RARA) immediately — different emergency pathway (ATRA±ATO).
- State blast threshold nuance: defining genetics can diagnose AML with <20% blasts in WHO 2022.
- End with allo-HCT indications tied to ELN risk and MRD.

EXAM PEARLS / HALLMARKS

- Open with 'genetics-first WHO 2022' — signals updated knowledge vs FAB-only answers.

2. WHO 2022 myeloid malignancies — overview map

Major WHO 2022 myeloid groups: clonal haematopoiesis / clonal cytopenia; myeloproliferative neoplasms (MPN: CML BCR::ABL1+, CNL, PV, ET, PMF, etc.); mastocytosis; myelodysplastic/myeloproliferative neoplasms (MDS/MPN); myelodysplastic neoplasms (MDS) with genetics-based entities (e.g., MDS with SF3B1, MDS with del(5q), MDS with biallelic TP53); AML with defining genetic abnormalities; AML defined by differentiation; secondary myeloid neoplasms; eosinophilic/TK-fusion disorders.

Key philosophy: integrate morphology, immunophenotype, cytogenetics, and NGS. ICC 2022 is a parallel system — mention awareness without derailing into a comparison essay unless asked.

- CML remains BCR::ABL1-defined MPN — not lumped into AML until blast phase criteria met.

- MDS vs AML boundary softened when AML-defining genetics present.
- Therapy-related and myelodysplasia-related AML flagged for adverse biology / HCT planning.

WHO 2022 myeloid diagnostic pathway

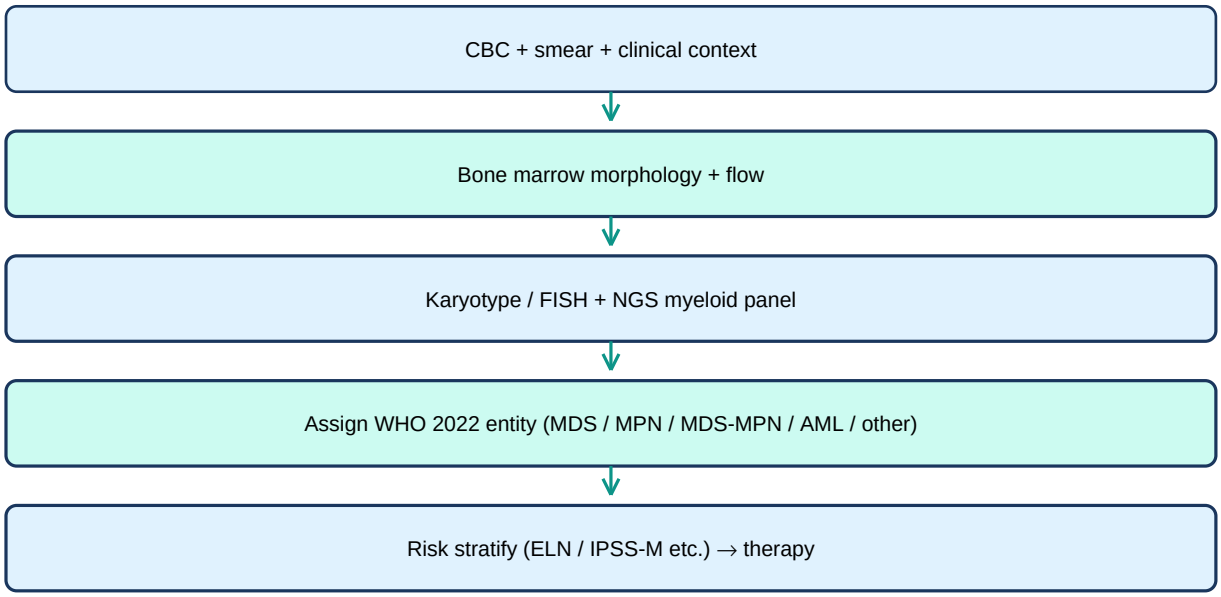


Figure: stepwise exam algorithm

3. AML — clinical presentation & urgent work-up

AML presents with marrow failure (anaemia, neutropenia, thrombocytopenia), leukocytosis/leukostasis, DIC (especially APL), infection, or extramedullary disease (gum, skin, CNS). Urgent priorities: stabilise airway/breathing, treat infection, cytoreduce if leukostasis, correct coagulopathy, and obtain diagnostic marrow before non-urgent cytoreductive chemo when feasible — but never delay ATRA if APL suspected.

- Labs: CBC, smear, PT/INR, aPTT, fibrinogen, D-dimer, chemistry, TLS labs, viral serologies, HLA typing early if HCT candidate.
- Imaging/LP only as indicated (CNS symptoms, monoblastic phenotype, relapse work-up).
- Echo/ECG before anthracyclines; PS, comorbidities, geriatric assessment for fit vs unfit.

First 48 hours in suspected AML

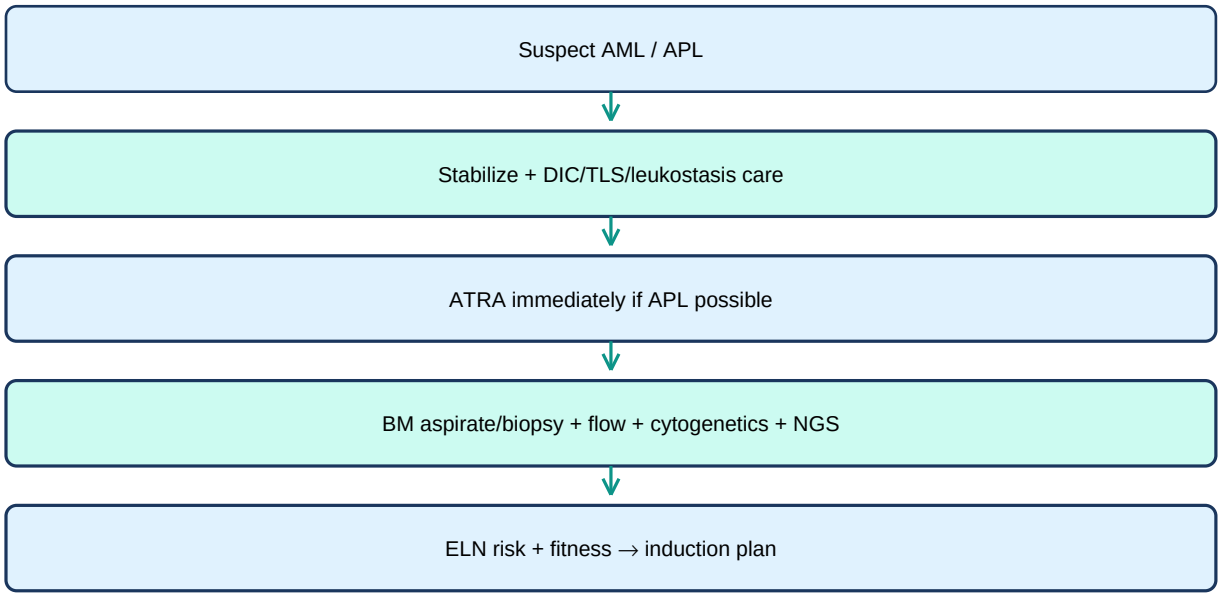


Figure: stepwise exam algorithm

4. Diagnostic investigations in detail

Morphology: blast percentage, Auer rods (myeloid), dysplasia. Flow cytometry: myeloid markers (MPO, CD13, CD33, CD117, CD34) vs monocytic (CD14, CD64) vs megakaryocytic; exclude mixed phenotype. Cytogenetics: karyotype essential for t(8;21), inv(16)/t(16;16), t(15;17), t(9;11), complex karyotype, etc.

WHO 2022 AML with defining genetic abnormalities includes (examples): PML::RARA; RUNX1::RUNX1T1; CBFB::MYH11; KMT2A rearrangements; DEK::NUP214; MECOM; NPM1; CEBPA bZIP; and others — learn the concept that genetics trump a rigid 20% blast cutoff for several entities.

Drug / regimen	Dose / schedule (exam pearls)
BM aspirate + biopsy	Morphology, iron stain as needed, trephine architecture
Flow immunophenotyping	Lineage assignment; baseline for MRD assays
Conventional karyotype	≥20 metaphases when possible; reflex FISH panels
Molecular NGS panel	NPM1, FLT3-ITD/TKD, IDH1/2, TP53, RUNX1, ASXL1, KRAS/NRAS, KIT, etc.
RT-PCR fusions	Rapid PML::RARA; CBF fusions; KMT2Ar as available

Table: doses and regimens

5. Molecular testing — what to order and why

At diagnosis order rapid FLT3 (ITD and TKD) because midostaurin/quizartinib timing matters with induction. NPM1, IDH1/2, TP53, and karyotype/FISH drive ELN risk and targeted options (ivosidenib/enasidenib/olutasidenib; menin inhibitors for NPM1/KMT2Ar). CBF AML: add KIT.

MRD tools: NPM1 RT-qPCR, multiparameter flow MRD, NGS MRD — ELN MRD guidance informs consolidation/HCT. Repeat NGS at relapse for targetable clones (FLT3 acquisition, menin pathway).

FLT3-ITD/TKD	Constitutive FLT3 signalling → proliferation; target with TKI
NPM1 mut	Cytoplasmic NPM1; HOX expression; menin dependence
KMT2A-r	Menin–KMT2A oncogenic transcription → menin inhibitors
IDH1/2 mut	2-HG oncometabolite → differentiation block; IDH inhibitors
TP53 / complex KY	Chemo resistance; adverse ELN; consider HCT / trials

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- FLT3 results should return within days — write 'do not delay sending rapid FLT3'.
- NPM1+ FLT3-ITD allelic ratio nuances appear in ELN — mention ITD ratio if space.

6. ELN 2022 risk stratification

ELN 2022 favourable: t(8;21)/RUNX1::RUNX1T1; inv(16)/CBFB::MYH11; mutated NPM1 without FLT3-ITD; bZIP in-frame CEBPA mutations. Intermediate: mainly cytogenetic/molecular findings not favourable or adverse (including FLT3-ITD with NPM1 per updated rules — quote current ELN table in exam). Adverse: complex karyotype, inv(3)/MECOM, –5/–7, TP53, ASXL1, RUNX1 (as sole), KMT2Ar (many), myelodysplasia-related genetics, etc.

Use ELN risk + MRD + patient fitness to decide consolidation intensity and allo-HCT in CR1. Adverse risk → HCT in CR1 if eligible; favourable → chemo consolidation if MRD-negative; intermediate → individualise with

MRD and donor availability.

- Always present ELN as favourable / intermediate / adverse with ≥2 example lesions each.
- Therapy-related and MRC-AML often behave adverse regardless of some genetics.
- Reassess risk at relapse with new genetics.

ELN 2022 → consolidative intent

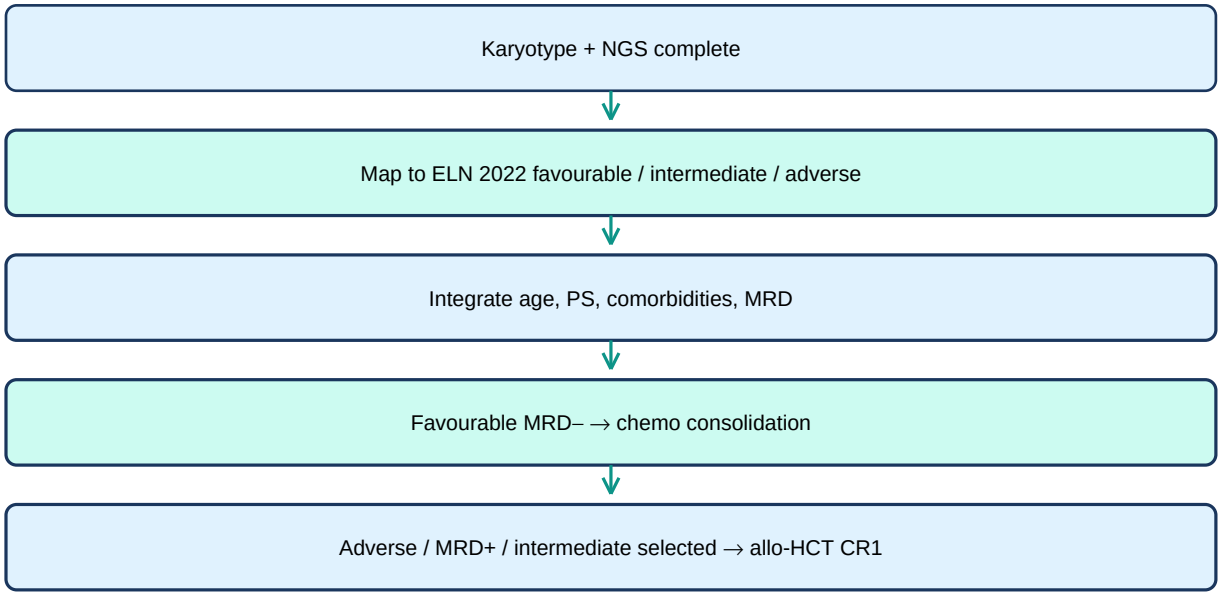


Figure: stepwise exam algorithm

7. Management — fit induction & consolidation

Fit induction standard: cytarabine + anthracycline ('7+3'). Add midostaurin for FLT3-mutated (RATIFY) or quizartinib for FLT3-ITD (QuANTUM-First) per label/eligibility. CBF AML may use GO (gemtuzumab ozogamicin) in selected protocols. APL: ATRA+ATO (± chemo) — separate section.

Consolidation: HiDAC cycles for favourable risk; bridge to allo-HCT for adverse/selected intermediate. Supportive care: antifungals, antibacterial prophylaxis per centre, transfusion, TLS monitoring, growth factors selectively.

Drug / regimen	Dose / schedule (exam pearls)
7+3 induction	Cytarabine 100–200 mg/m² CIV D1–7 + daunorubicin 60 mg/m² D1–3 (or idarubicin 12 mg/m²)
Midostaurin (FLT3)	50 mg PO BID D8–21 each induction/consolidation cycle (RATIFY)
Quizartinib (FLT3-ITD)	Per QuANTUM-First with chemo — monitor QTc
HiDAC consolidation	Cytarabine 1–3 g/m² Q12h on D1,3,5 (age/centre-adjusted)
GO (selected CBF/CD33)	Fractionated GO with chemo per protocol — VOD risk counselling

Table: doses and regimens

EXAM PEARLS / HALLMARKS

- Write exact 7+3 doses once — free marks.
- Anthracycline cardiotoxicity: baseline EF; lifetime dose tracking.

8. Unfit / older AML — HMA + venetoclax backbone

VIALE-A established azacitidine + venetoclax vs azacitidine in unfit newly diagnosed AML: median OS 14.7 vs 9.6 months; higher composite CR rates. Decitabine + venetoclax is an alternative HMA partner. Ramp venetoclax with TLS prophylaxis; adjust for azole antifungals (CYP3A).

IDH1-mutated unfit: ivosidenib + azacitidine (AGILE OS benefit) or ivosidenib monotherapy in selected. NCCN 2026 reinforces HMA+ven after cytarabine-based induction failure and as frontline unfit standard.

Trial	Key result / exam takeaway
VIALE-A	Aza + ven vs aza: mOS 14.7 vs 9.6 mo — unfit ND-AML standard
AGILE	Ivosidenib + aza vs aza in IDH1 mut — EFS/OS benefit
RATIFY	7+3 + midostaurin FLT3 mut — OS benefit
QuANTUM-First	Quizartinib + chemo FLT3-ITD — OS benefit
ADMIRAL	Gilteritinib vs salvage chemo R/R FLT3 — OS benefit

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Azacitidine	75 mg/m² SC/IV D1–7 (or D1–5/8–9 schedules) Q28D
Venetoclax (AML)	Ramp 100 → 200 → 400 mg daily with HMA; hold for cytopenias per label
Ivosidenib	500 mg PO daily (IDH1); differentiation syndrome watch
Enasidenib	100 mg PO daily (IDH2 R/R context)

Table: doses and regimens

9. Targeted agents — FLT3, IDH, menin (NPM1 / KMT2A)

FLT3: midostaurin/quizartinib with induction; gilteritinib in R/R FLT3 (ADMIRAL). Monitor QTc, differentiation syndrome rare, interactions. IDH inhibitors: differentiation syndrome (treat like APL DS with steroids), QTc (ivosidenib), bilirubin (enasidenib).

Menin inhibitors: revumenib for KMT2A-rearranged R/R AML (and related indications); ziftomenib for R/R NPM1-mutated AML — both appear in NCCN AML v5.2026. These exploit dependence on menin–KMT2A transcriptional complexes driving HOX/MEIS1.

- NCCN 2026 also lists intensive chemo + ven combinations (CLIA+ven; FLAG-IDA+ven) as Cat 2B options.
- Olutasidenib Cat 2B IDH1 option if not candidate for preferred/ivosidenib.
- Always send patients to allo-HCT evaluation in parallel with targeted salvage when curative intent remains.

Menin–KMT2A complex	Drives leukemogenic HOX program in KMT2Ar / NPM1
Revumenib	Menin inhibitor — KMT2Ar R/R AML (NCCN listed)
Ziftomenib	Menin inhibitor — R/R NPM1-mut AML (NCCN Cat 2A)
Resistance themes	Menin mutations / pathway reactivation — trial space

Figure: mechanism / pathway map

10. Relapsed/refractory AML algorithm

Confirm relapse with marrow ± extramedullary assessment; repeat cytogenetics/NGS. Choose salvage by prior therapy, fitness, and targets: gilteritinib (FLT3), IDH inhibitors, menin inhibitors (NPM1/KMT2Ar), HMA+ven if not refractory to it, intensive salvage (FLAG-IDA ± ven), clinical trials, then HCT if CR2 achievable.

R/R AML decision tree (include NCCN 2026 agents)

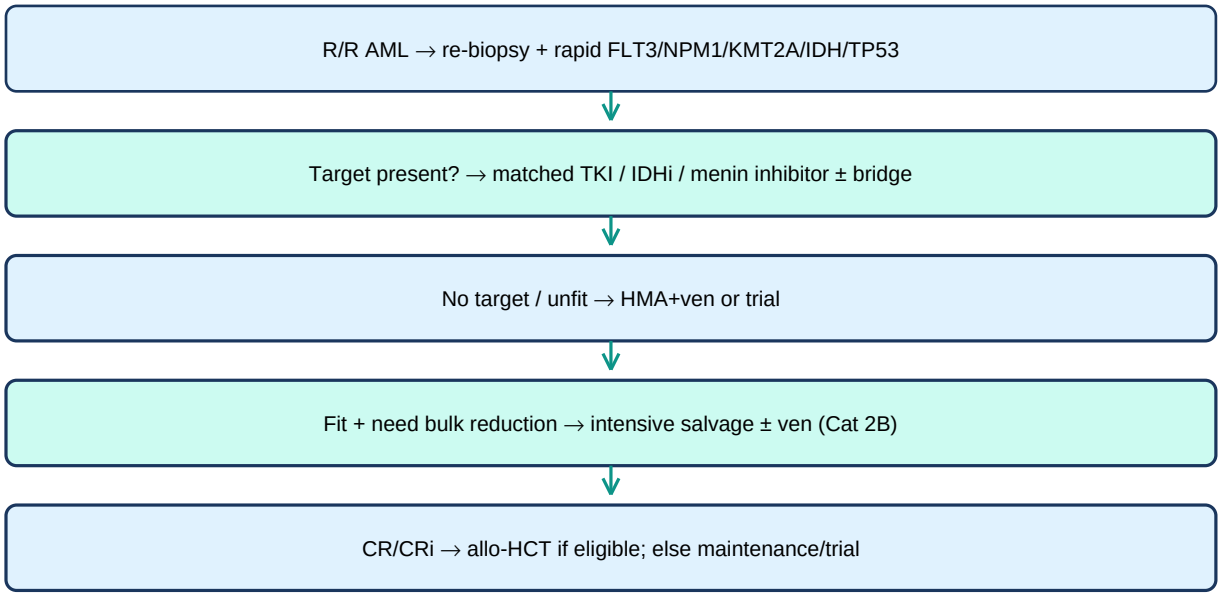


Figure: stepwise exam algorithm

Trial	Key result / exam takeaway
ADMIRAL	Gilteritinib R/R FLT3 — mOS ~9.3 vs 5.6 mo vs salvage chemo
Menin programs	Revumenib KMT2Ar responses; ziftomenib NPM1 R/R — NCCN 2026
Bridging concept	Targeted salvage → HCT consolidation whenever feasible

Table: landmark clinical trials

11. APL — do not mix into generic AML

APL is a medical emergency defined by PML::RARA. Start ATRA on suspicion before genetic confirmation if coagulopathy/compatible morphology. Low/intermediate risk: ATRA+ATO (Lo-Coco / AML17 paradigms); high-risk: add chemo/GO per protocol. Manage DIC aggressively; watch differentiation syndrome (dexamethasone).

Drug / regimen	Dose / schedule (exam pearls)
ATRA	45 mg/m²/day PO divided until CR (protocols vary)
ATO	0.15 mg/kg IV daily per induction/consolidation schedules
Supportive	Aggressive FFP/cryoprecipitate for DIC; platelets keep high thresholds

Table: doses and regimens

EXAM PEARLS / HALLMARKS

- Exam killer omission: forgetting immediate ATRA when APL suspected.
- ATO toxicity: QTc prolongation — replete K/Mg; serial ECG.

12. NCCN AML v5.2026 updates box

Examiners increasingly expect named 2026 agents. Place this box near R/R and unfit sections.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Acute Myeloid Leukemia v5.2026
- Ziftomenib Category 2A for relapsed/refractory NPM1-mutated AML.
- Revumenib listed for R/R AML in menin inhibitor / KMT2A rearrangement context.
- HMA (azacitidine/decitabine) + venetoclax pathway reinforced for unfit patients and after cytarabine-based induction failure.

- CLIA + venetoclax and FLAG-IDA + venetoclax Category 2B intensive options.
- Olutasidenib Category 2B IDH1 option if not candidate for preferred regimens / ivosidenib.
- Continue FLT3 inhibitor integration (midostaurin/quizartinib induction; gilteritinib R/R) per mutation.

EXAM PEARLS / HALLMARKS

- Spell ziftomenib ↔ NPM1 and revumenib ↔ KMT2Ar explicitly — easy discriminator marks.

13. Supportive care, infections & HCT principles

TLS prevention, transfusion support, menstrual suppression, tumour fever vs infection work-up, antifungal prophylaxis in induction, and dental/line care matter for OS. Allo-HCT: matched sibling/unrelated/haplo with PTCY — indicate for adverse ELN, persistent MRD, secondary AML, and most CR2. Vaccinations and late-effects follow-up post-HCT.

- Leukostasis: hydroxyurea ± leukapheresis in selected hyperleukocytosis with symptoms.
- Avoid azole–venetoclax interaction errors — dose-adjust ven.
- Fertility counselling before induction when feasible.

14. Summary — wall-chart answer & pearls

Final page: WHO genetics-first map → AML work-up → ELN 2022 → fit 7+3±FLT3i vs unfit HMA+ven → targets (FLT3/IDH/menin) → HCT → NCCN 2026 names.

- WHO 2022: genetics can define AML without rigid 20% blasts.
- ELN 2022 drives HCT timing.
- VIALE-A and RATIFY/QuANTUM-First/ADMIRAL are core numbers.
- NCCN 2026: ziftomenib NPM1 R/R; revumenib; HMA+ven; FLT3 pathway.

One-page AML wall chart

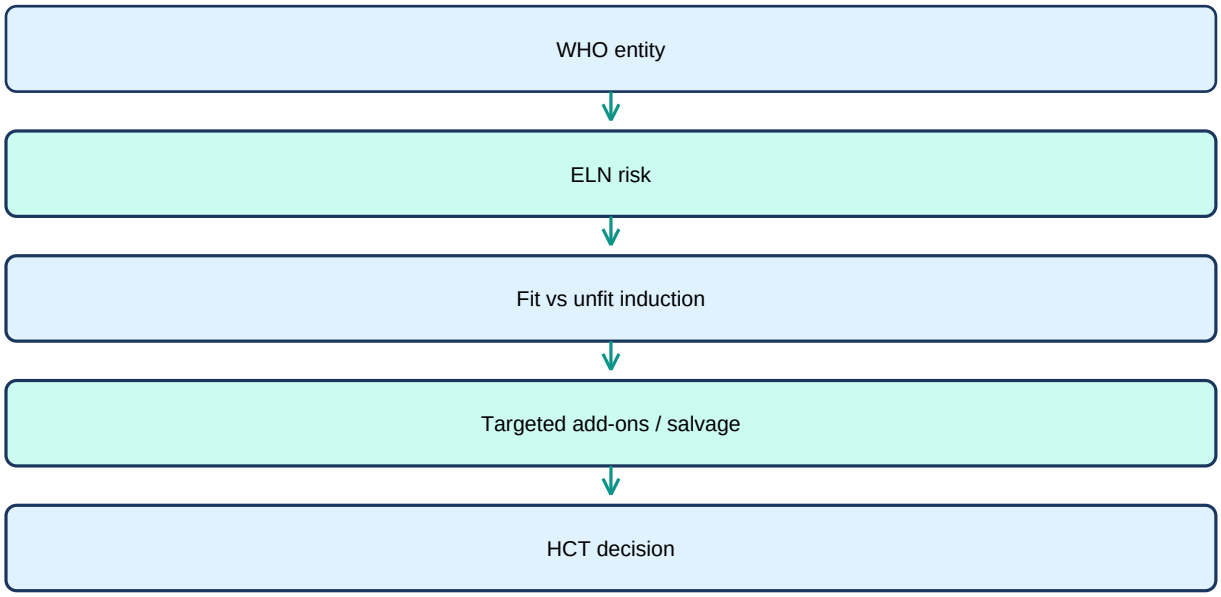


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- APL out of the gate with ATRA.
- Rapid FLT3 at diagnosis every time.
- VIALE-A mOS 14.7 vs 9.6 mo.

• Menin inhibitors are the 2026 'recent advances' punchline.

• MRD-guided consolidation is modern ELN practice — say it aloud in the answer.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

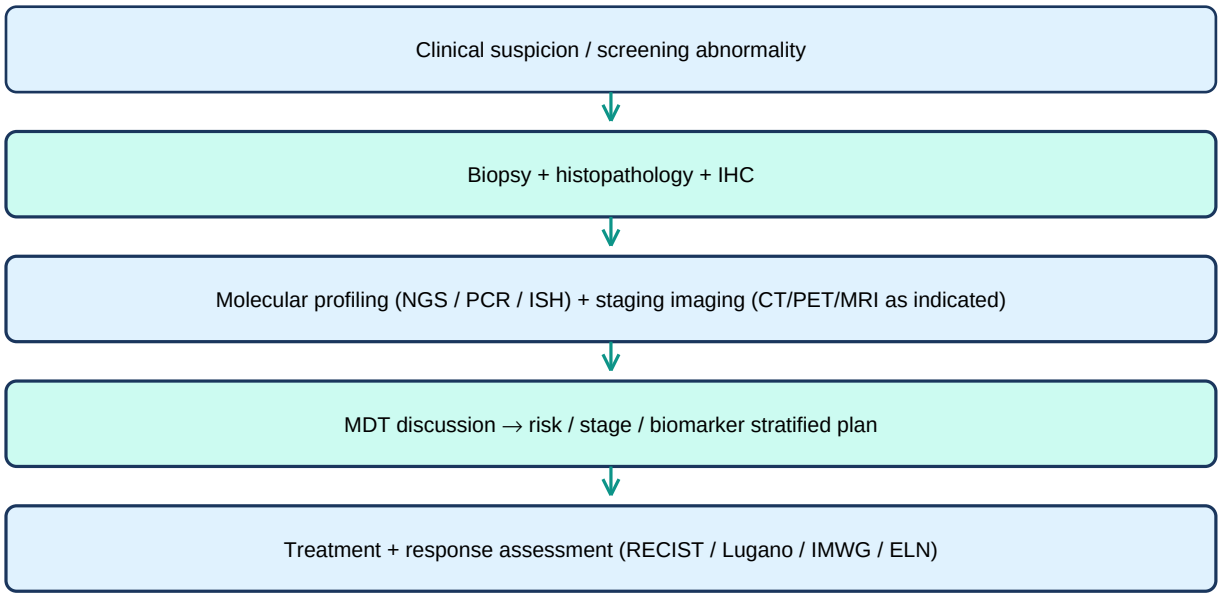


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.

• Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).

• Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

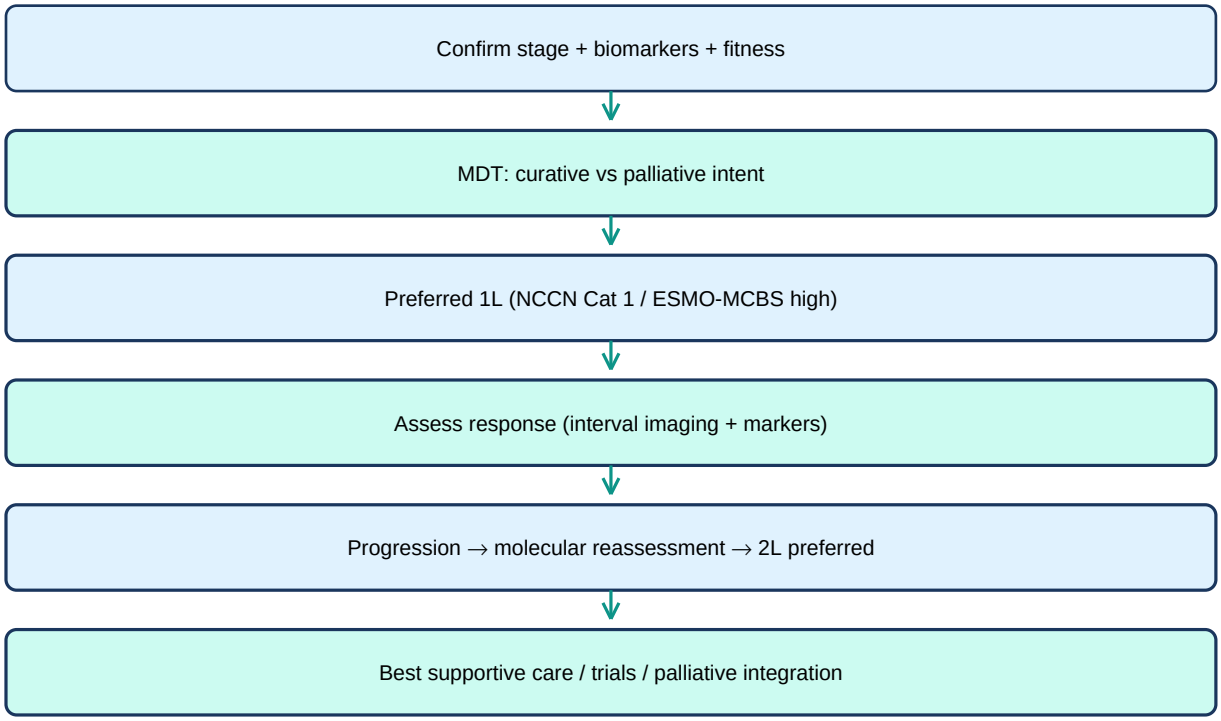


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Venetoclax (AML)	Ramp 100→200→400 mg with HMA; tumor lysis prophylaxis
Venetoclax (CLL)	5-week ramp to 400 mg with anti-CD20 (CLL14/MURANO)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

• Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.

• Distinguish statistical vs clinical significance; note crossover and subgroup caveats.

• For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
RATIFY	Midostaurin + 7+3 FLT3
ADMIRAL	Gilteritinib R/R FLT3
VIALE-A	Azacitidine + venetoclax unfit AML
QuANTUM-First	Quizartinib + chemo FLT3-ITD
AGILE	Ivosidenib + aza IDH1

Brummett / menin inhibitor programs	Revumenib KMT2Ar; ziftomenib NPM1 — NCCN 2026 R/R

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

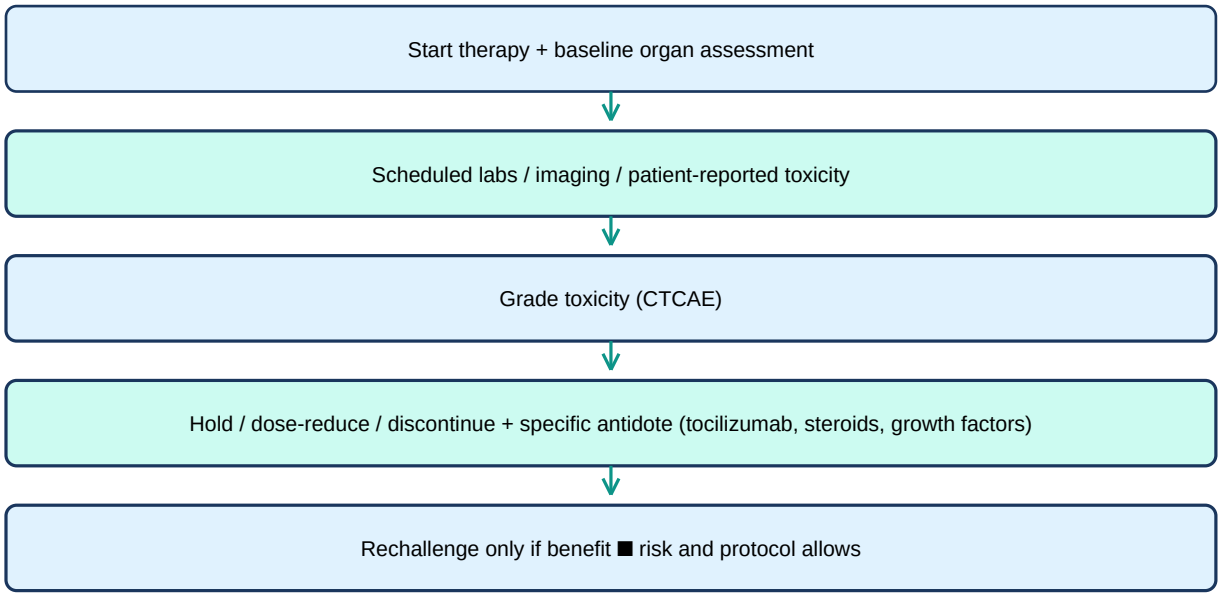


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Acute Myeloid Leukemia v5.2026
- Ziftomenib Cat 2A for R/R NPM1-mutated AML.
- CLIA + venetoclax and FLAG-IDA + venetoclax Cat 2B intensive options.
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- Olutasidenib Cat 2B IDH1 option if not candidate for preferred / ivosidenib.
- Revumenib listed for R/R (menin inhibitor / KMT2A rearrangements context).

EXAM PEARLS / HALLMARKS

- NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).
- Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

ELN 2022 risk (favorable/intermediate/adverse) integrates cytogenetics and mutations (NPM1, FLT3-ITD allelic ratio historically, TP53, MECOM, complex karyotype, etc.). WHO 2022 / ICC classifications emphasize genetically defined entities.

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

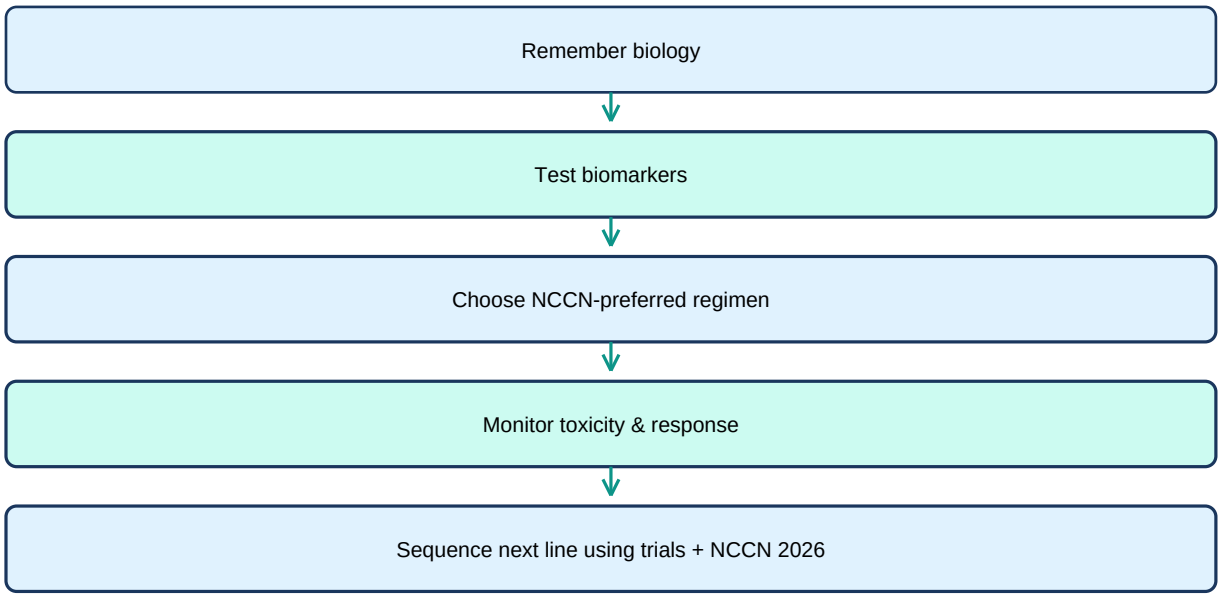


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say ‘NCCN 2026’ explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Venetoclax (AML)	Ramp 100→200→400 mg with HMA; tumor lysis prophylaxis
Venetoclax (CLL)	5-week ramp to 400 mg with anti-CD20 (CLL14/MURANO)
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Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.

• PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

• ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

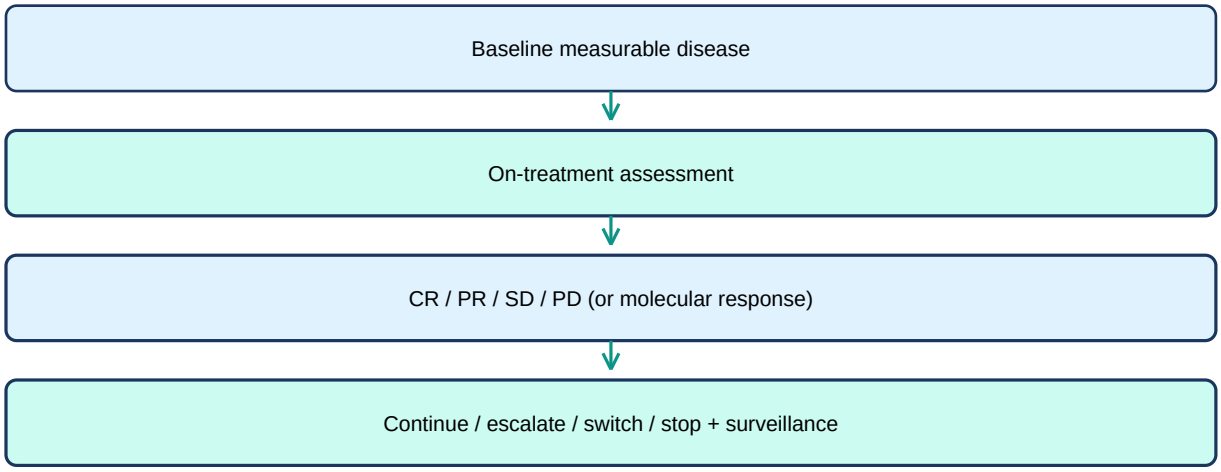


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

• Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.

• Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
RATIFY	Midostaurin + 7+3 FLT3
ADMIRAL	Gilteritinib R/R FLT3
VIALE-A	Azacitidine + venetoclax unfit AML
QuANTUM-First	Quizartinib + chemo FLT3-ITD
AGILE	Ivosidenib + aza IDH1
Brummett / menin inhibitor programs	Revumenib KMT2Ar; ziftomenib NPM1 — NCCN 2026 R/R

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

• Document MDT decisions; clinical trial referral when standard options exhausted.

• Counsel financial toxicity transparently; prioritize highest-value Category 1 options.

• Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

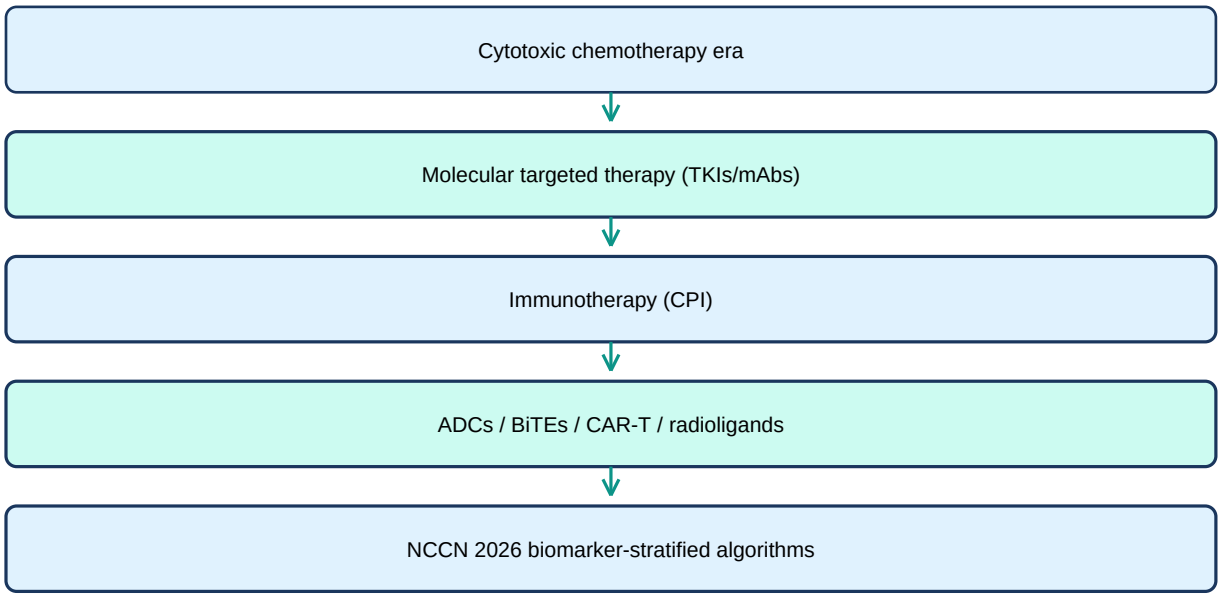


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

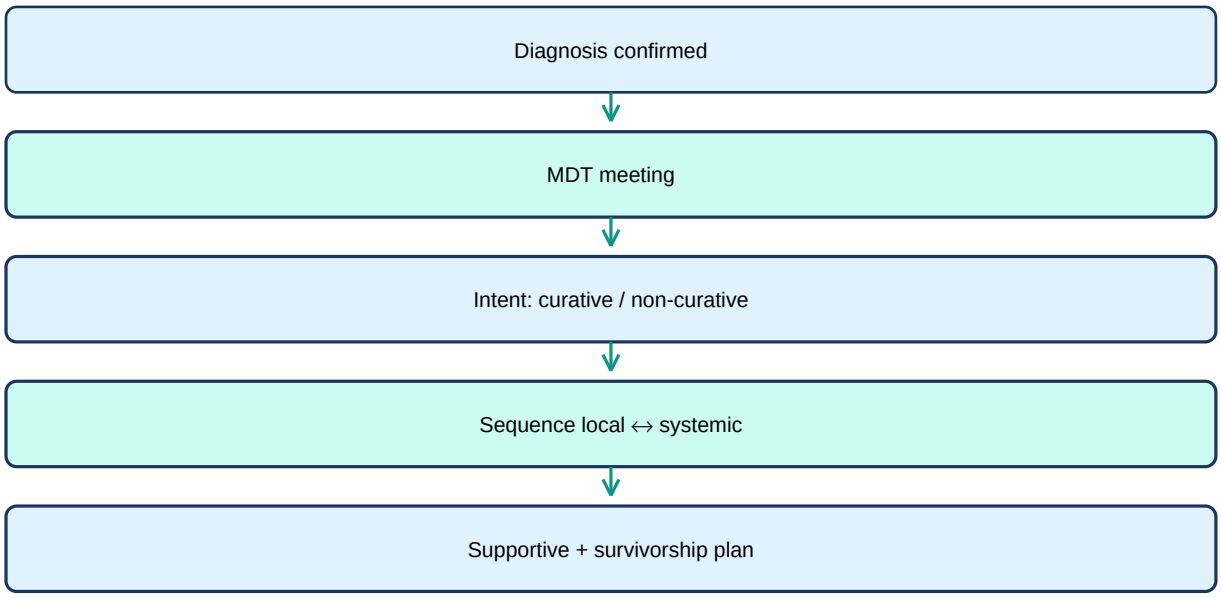


Figure: stepwise exam algorithm

E. AML / myeloid deep notes

Fit induction: 7+3 ± FLT3 inhibitor (midostaurin/quizartinib per mutation). Unfit: HMA+venetoclax (VIALE-A).

APL separate: ATRA+ATO (±ida) with meticulous coagulopathy management — highest curability.

ELN 2022 risk guides allograft timing. MRD (flow/NGS) increasingly guides consolidation.

NCCN 2026: ziftomenib for R/R NPM1; revumenib for R/R (menin); olutasidenib IDH1 option; intensive chemo+venetoclax combinations Cat 2B.

Clinical algorithm

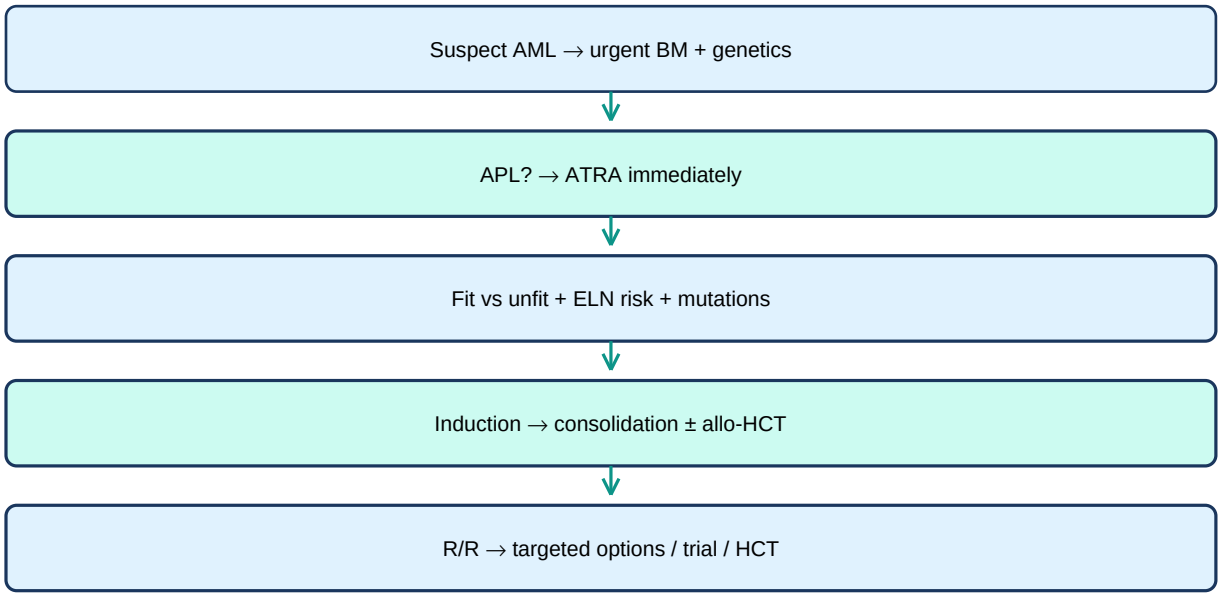


Figure: stepwise exam algorithm

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

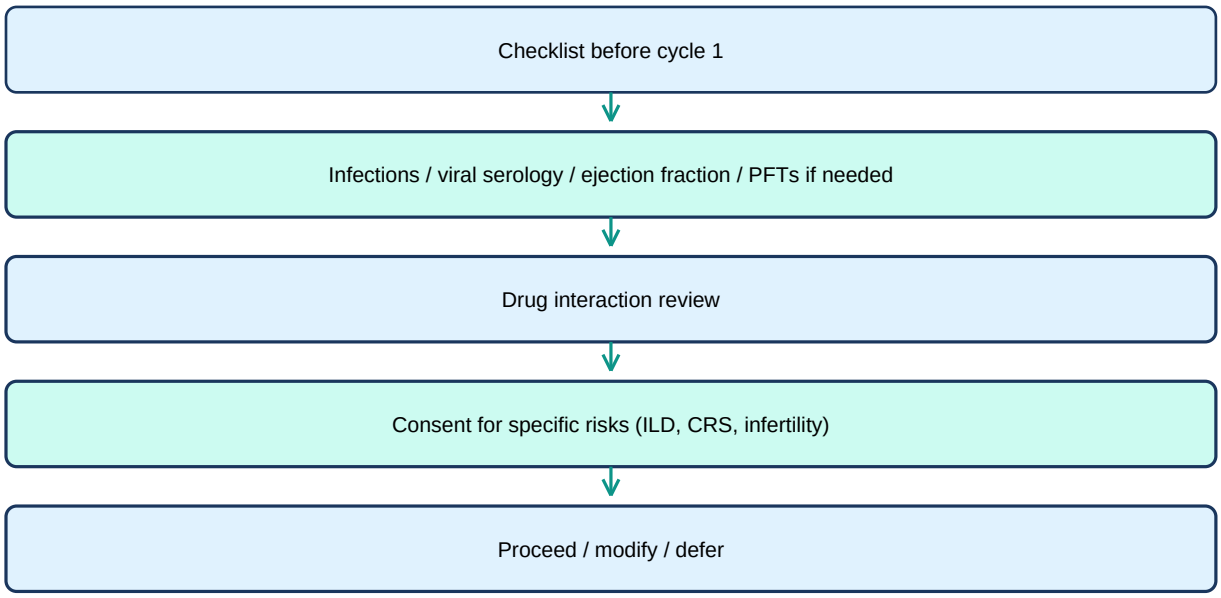


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write 'NCCN 2026' in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
RATIFY	Midostaurin + 7+3 FLT3
ADMIRAL	Gilteritinib R/R FLT3
VIALE-A	Azacitidine + venetoclax unfit AML
QuANTUM-First	Quizartinib + chemo FLT3-ITD
AGILE	Ivosidenib + aza IDH1
Brummett / menin inhibitor programs	Revumenib KMT2Ar; ziftomenib NPM1 — NCCN 2026 R/R

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Venetoclax (AML)	Ramp 100→200→400 mg with HMA; tumor lysis prophylaxis
Venetoclax (CLL)	5-week ramp to 400 mg with anti-CD20 (CLL14/MURANO)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Acute Myeloid Leukemia v5.2026
- Ziftomenib Cat 2A for R/R NPM1-mutated AML.
- CLIA + venetoclax and FLAG-IDA + venetoclax Cat 2B intensive options.
- HMA (aza/decitabine) + venetoclax pathway reinforced for unfit / after cytarabine-based induction failure.
- Olutasidenib Cat 2B IDH1 option if not candidate for preferred / ivosidenib.
- Revumenib listed for R/R (menin inhibitor / KMT2A rearrangements context).

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one 'if progression then…' resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

NCCN 2026 / 2027 UPDATES (include in answer)

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- Ziftomenib Cat 2A for R/R NPM1-mutated AML.
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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.